

JAWAHARLAL NEHRU UNIVERSITY

SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

Final

M.C.A. (Monsoon Semester-2022)

Course Name: Computer Architecture

Maximum Marks: 60

Course Code: CS 405

Total Time: 3 hours

- Q1.** a) What is the difference between X86 and ARM type of Computer Architecture?
b) What are the major factors (performance parameters) that can help in improving the CPU performance in a Computer System?
c) Three enhancements with following speedups are proposed for new architecture:
Speedup₁=25; Speedup₂=15; Speedup₃=10; only one enhancement is useable at time
(i) If enhancement 1 and 2 are each usable for 21% of the time, what fraction of the time must enhancement 3 be used to achieve overall speed up of 15?
(ii) Assume the enhancement can be used 20%, 35% and 18% of the time for enhancement 1, 2, 3 respectively. For what fraction of the reduced execution time no enhancement is used?

(3+3+4)

Q2. a) A non pipeline processor has a clock frequency of 300 M Hz and an average CPI of 4. A better processor Y is designed with a 5 stage instruction pipeline. However to latch delay and clock slew the clock rate of Y is 250 M Hz. If a program consisting of 5000 instructions are executed by both the processors, what will be the speed up of processor Y with respect to processor X.

- b) For the following reservation table
(i) What are the forbidden latencies
(ii) Show the state transition diagram
(iii) List all the simple and greedy cycles
(iv) Determine the optimum MAL
(v) Determine the pipeline throughput with $\Delta = 10\text{ns}$

	1	2	3	4	5	6
S1	X		X		X	
S2		X		X		X
S3			X		X	

	1	2	3	4	5
S1	Y			Y	Y
S2		Y	Y	Y	
S3	Y	Y		Y	

(3+7)

Q3. a) What is the difference between 1 cell SRAM and DRAM. Which type of memory is used at various levels of memory Hierarchy Design and why?

b) Consider the pipeline stage with

	Load	Store	R type	Branch	Jump
Instruction Mix	20%	15%	40%	15%	10%

Assuming memory access is 200 ps, ALU operation is 50 ps, register file read/write is 20 ps, half of the load instruction is immediately followed by an instruction that uses the results of branch delay on mis-prediction on one clock cycle and $\frac{1}{4}$ of the branches are mis-predicted. Jumps pays one full clock cycle of delay. Calculate CPI, cycle time and target address in ID stage to fetch the next instruction so that average time in pipe is 2 clock cycle. (For pipelining average load =1.5 clock cycles; store =1 load cycle; R-type =1clock cycle; jumps =2 clock cycle; average branch =1.5 clock cycles)

(4 +6)

Q4. What are the different ways to improve cache performance.

(b) Consider a CPU with an average CPI as 2.2. Assume the instruction mix is ALU-50%, Load-15%, Branch-35%. Assume the cache miss rate of 2.1% and miss penalty of 50 cycles ($=t_{MM}$). Calculate the effective CPI for a unified L1 cache, using write back and write allocate, with the cache block being dirty is 12%.

(3 +7)

Q5. a) What is Flynn classification for Computer Systems?

b) Discuss vector processing. Consider the multiplication of two 30 X 30 matrices using vector processor. How many product terms are there in the inner product and how many inner products must be evaluated?. How many multiply add operations are needed to calculate product matrix.

c) Suppose start up time for vector multiply operation is 15 clock cycles. After start up, initiation rate is 1.5 per clock cycle. What will be the number of clock cycles required per result for a 64 element vector?.

(3 +5+2)

Q6. a) Explain various types of hazards in the pipelining technique?. What is stalling?.

b) A weather forecasting computation requires 300 billion floating point operations. The problem is processed in a Super computer that can perform 100 megaflops. How long will it take to perform these calculations?

c) Explain arithmetic pipelining.

(4 +3+3)

JAWAHARLAL NEHRU UNIVERSITY
SCHOOL OF COMPUTER AND SYSTEMS SCIENCES
Final Exam

M.CA. (Winter Semester-2023)

Shreya Pal

Course Name: Computer Architecture

Maximum Marks: 60

Course Code: CS 106

Total Time: 3.0 hours

Q1. (a) What is the difference between X86 and ARM type of Computer Architecture?

(b) Explain the concept of parallelism in Computer System.

(c) Suppose a program is running on a machine with the following instruction types, CPI values and the frequencies of occurrence. The CPU designer gives two options i) reduce CPI instruction type A to 1.2 and ii) reduce CPI instruction type B to 1.5. Which one is better?

Type	CPI	Frequency
A	1.5	50%
B	2.5	30%
C	2.1	20%

(3+2+5)

Q2. (a) What is flash memory and which part of memory hierarchy it is placed? Explain the term locality of reference.

(b) What conclusion can one draw from the following observations in terms of computing performance

Type	CPU A (in Sec.)	CPU B (in Sec.)	CPU C (in Sec.)
Program A	60	40	30
Program B	510	220	90
Total Time	570	260	120

(4+6)

Q3. (a) What is a Vector Processor?

(b) The great advantage of pipelining is the performance improvement with a shorter clock time.

Element	Register Clk to q	Register Set Up	Mux.	ALU	Mem. Read	Mem. Write	Register File Read	Register File Set Up
Parameter	t clk-to-q	t Setup	t Mux.	t ALU	t MEM read	t MEM write	t RF Read	t RF setup
Delay (ps)	30	20	25	200	250	200	150	5

i) What was the clock time and frequency of a single cycle CPU?

ii) What is the clock time and frequency of a pipelined CPU?

iii) What is the speed-up?

(4+6)

Q4. (a) What are the guiding parameters for defining the memory Hierarchy?

(b) Consider two different types of memory hierarchy. In the first case, consider a three level memory hierarchy with separate instruction and data cache in level 1 while the main memory is at level 2. In the second case, consider a three level memory hierarchy with separate instruction and data cache in level 1 and level 2 respectively. The main memory is at level 3.

Calculate the average access time in both types of memory hierarchy for the following parameters:

- i) Clock cycle is 2 ns
- ii) Miss cycle is 10 clock cycle
- iii) 2 percent of instruction is not found in Instruction cache
- iv) 9 percent of instruction is not found in Data cache
- v) 25 percent of the total memory access for the data
- vi) Cache Access Time (including hit detection) is 1 clock cycle.

Which one of the memory configuration is better?

(2+8)

Q5. (a) What is the difference between Deep and Shallow pipelining technique?

(b) For the following reservation table

- (i) What are the forbidden latencies
- (ii) Show the state transition diagram
- (iii) List all the simple and greedy cycles
- (iv) Determine the optimum MAL
- (v) Determine the pipeline throughput with $\tau=20\text{ns}$

	1	2	3	4	5	6
S1	X		X	X	X	
S2		X		X		X
S3		X	X		X	

	1	2	3	4	5
S1	Y		Y	Y	Y
S2		Y		Y	
S3	Y	Y			

(2+8)

Q6. (a) What are the different ways for cache optimization techniques.

(b) Suppose that in 1000 memory references there are 60 misses in L-1 cache are 30 misses in L-2 cache and cache are 10 misses in L-3 cache. What are various miss rates?

Assume that Miss Penalty $L-2$ is 180 clock cycles, Hit time $L-1$ is 1 clock cycle, Hit time $L-2$ is 6 clock cycle, Miss Penalty $L-3$ is 90 clock cycles, Hit time $L-3$ is 12 clock cycle

What will be the average memory access Time?. Ignore impact of write.

(4+6)

Aditya

JAWAHARLAL NEHRU UNIVERSITY
SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

Final Exam

M.CA. (Winter Semester-2024)

Course Name: Computer Architecture

Course Code: CS 106

Maximum Marks: 60

Total Time: 3.0 hours

- Q1. (a) Why RISC Architecture is preferred over X86 in the present day smart phones?
(b) What are the factors affecting the performance of Computer Architecture
(c) Determine the normalized execution time using arithmetic mean and the geometric mean

Type	CPU A	CPU B	CPU C
Program P1	5	15	25
Program P2	250	100	15
Total Time			

Which one is preferred and why

(3+3+4)

- Q2. (a) Explain the concept of parallelism or pipelining in context to CPU and GPU.
(b) A quad core processor could speed up a computer by a factor of 4 but this rarely happens. Compute the percentage of program execution that needs to be distributed across all 4 cores to achieve an overall speedup of 3. Of 2. Of 1.5. Of 1.25.

(c) You have a computing task that can be divided into three parts:

1. Part A: 50% of the task can be fully parallelized.
2. Part B: 30% of the task can be parallelized.
3. Part C: 20% of the task is inherently serial and cannot be parallelized.

You want to calculate the overall speedup when using 16 processors for Part A and 8 processors for Part B. Part C remains serial.

(3+4+5)

Q3. . (a) What is the difference between cache memory, flash memory and virtual? Explain the term locality of reference and spatial locality.

(b) Explain about the type of memory used in primary memories.

(c) Write an assembly program to explain different types of addressing modes in 8085 microprocessor.

(5+2+3)

Q4. (a) What are the basic cache optimization techniques?

(b) Calculate the average access time in both types of memory hierarchy for the following parameters:

- i) Clock cycle is 5 ns
- ii) Miss cycle is 10 clock cycle
- iii) 5 percent of instruction is not found in Instruction cache
- iv) 10 percent of instruction is not found in Data cache
- v) 25 percent of the total memory access for the data
- vi) Cache Access Time (including hit detection) is 1 clock cycle.

Which one of the memory configuration is better?

(3+6)

Q5. (a) What is Dynamic and a Non-linear pipelining technique?

(b) For the Multiplier and Addition reservation table

- (i) Determine all the forbidden latencies
- (ii) Show the state transition diagram
- (iii) Determine the optimum MAL
- (v) Determine the pipeline throughput with $\Delta = 20\text{ns}$
- (vi) What is the efficiency of the pipeline technique.

(3+7)

Q6. (a) Explain the features of vector processor Architecture?. What are the different Taxonomy of Parallel Architectures.

(b) What is the advantage of using multi core?. Explain the difference between loosely coupled and tightly coupled Microprocessor.

(4+4)

JAWAHARLAL NEHRU UNIVERSITY

SCHOOL OF COMPUTER AND SYSTEMS SCIENCES

Final Exam

M.CA. (Winter Semester-2025)

Course Name: Computer Architecture

Course Code: CS 106

Maximum Marks: 40

Total Time: 3 hour

Q1. a) Implement Universal gates using transistors.

b) What is the role of decoder and explain its role in a Computer system

c) What are the additional addressing modes in 8086 micro-processor. Write an assembly language program in 8086 to perform an XOR operation. (3+2+5)

Q2. (a) Compute the total cycles required using the Harvard and Von Neumann Architecture where a loop adds 10 numbers stored in an array and stores the sum in memory. Instructions per iteration has Load number, add to accumulator and store accumulator. The Memory access is one cycle per fetch/store and total iterations are 10.

b) A program has 1 million instructions to be executed using either CISC processor whose CPI = 1.1, Clock = 2.0 GHz or RISC processor whose CPI = 2, Clock = 1.5 GHz. Which one is faster?

c) Write an assembly language program to show the difference between Reduced Instruction set and Complex Instruction set. (4+3+3)

Q3. (a) Consider a CPU with an average CPI as 2.5. Assume the instruction mix is ALU-35%, Load-15%, Store-20% Branch-30%. Assume the cache miss rate of 15% and miss penalty of 120 cycles ($=t_{MM}$). Calculate the effective CPI for a unified L1 cache.

(b) A computing task that can be divided into three parts:

1. Part A: 40% of the task can be fully parallelized.
2. Part B: 45% of the task can be parallelized.
3. Part C: 15% of the task is inherently serial and cannot be parallelized.

Calculate the overall speedup when using 12 processors for Part A and 4 processors for Part B. Part C remains serial. (5+5)

Q4. (a) Explain the working of one unit of SRAM cell?

(b) What parameters play a pivotal role in deciding the memory hierarchy. What is the role of cache and virtual memory?

c) A computer has the following cache memory system: Cache size = 16 KB, Block size = 64 bytes, 4-way set associative cache, Main memory address = 16 bits. How many sets are in the cache?. How many bits are used for block offset, set index, and tag?

(3+3+4)

Q5. (a) What is the difference between static and dynamic pipelining techniques?

(b) Explain the different types of pipelining hazards?

©) What is the difference between linear and non-linear pipelining.

b) For the following reservation table

(i) What are the forbidden latencies

(ii) Show the state transition diagram

(iii) List all the simple and greedy cycles

(iv) Determine the optimal constant lattice cycle and MAL

(v) Determine the pipeline throughput with $\tau=15\text{ns}$

	1	2	3	4	5	6
S1	X		X			X
S2		X	X	X		
S3	X				X	

	1	2	3	4	5
S1	X			X	X
S2	X		X		
S3		X		X	

(2+3+5)

Q6. (a) Write is a Vector processor?.

(b) Explain the difference between Tightly coupled and Loosely coupled microprocessor. Which one is preferred is preferred and why?

(c) What are the advantages and disadvantages of hardwired vs microprogrammed control units?

(2+4+4)